Examiners' Report
Lead Examiner Feedback

January 2023

Pearson BTEC Nationals
In Applied Science (31617H)
Unit 1: Principles and Applications of Science I

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Grade Boundaries

What is a grade boundary?

A grade boundary is where we set the level of achievement required to obtain a certain grade for the externally assessed unit. We set grade boundaries for each grade, at Distinction, Merit and Pass.

Setting grade boundaries

When we set grade boundaries, we look at the performance of every learner who took the external assessment. When we can see the full picture of performance, our experts are then able to decide where best to place the grade boundaries – this means that they decide what the lowest possible mark is for a particular grade.

When our experts set the grade boundaries, they make sure that learners receive grades which reflect their ability. Awarding grade boundaries is conducted to ensure learners achieve the grade they deserve to achieve, irrespective of variation in the external assessment.

Variations in external assessments

Each external assessment we set asks different questions and may assess different parts of the unit content outlined in the specification. It would be unfair to learners if we set the same grade boundaries for each assessment, because then it would not take accessibility into account.

Grade boundaries for this, and all other papers, are on the website via this link

Introduction

Biology

SECTION A: STRUCTURES AND FUNCTIONS OF CELLS AND TISSUES

Medical professionals need to understand the structure and workings of cells. They build on this knowledge to understand how the body stays healthy as well as the symptoms and causes of some diseases. This allows them to diagnose and treat illnesses. The study of bacterial prokaryotic cells gives an understanding of how some other diseases are caused and can be treated.

Introduction to the Overall Performance of the Unit

Biology

SECTION A: STRUCTURES AND FUNCTIONS OF CELLS AND TISSUES

Learners should show that they understand the relationship between the structure and function of cells and tissues. Learners still struggle to recall definitions, but the understanding of the command verbs has improved. It is beneficial for learners to appreciate the requirement of each of the command verbs so that they can target their response and provide appropriate credit worthy answers. Explain is the command verb which seems least understood, especially for a 2-mark question. Learners tend to make an identification point without a linked expansion for full marks, or learners give a number of points, suggesting an imbalance of time spent on short questions.

Time management and the depth of knowledge expected on some key areas of the specification seem to be the biggest issue for learners. The additional guidance document exemplifies the expectations of the specification. Centres need to fully prepare learners for the exam by practising exam technique, especially in relation to reading the question carefully and not repeating the stem of the question. Learners should also be taught that when they have answered the question to reread their response in order to ensure that the question set has been addressed in the answer they have given, and that they have used appropriate scientific knowledge and vocabulary.

The specification and sample assessment materials (SAMs) are located on the BTEC National qualification webpage, located here.

Introduction

Chemistry

SECTION B: PERIODICITY AND PROPERTIES OF ELEMENTS

It was pleasing to see that, once again, learners appear to be becoming proficient in these papers with fewer blank spaces seen and many more learners achieving marks across the whole of the paper. Centres should continue to ensure that their learners are aware of the content being tested using the additional guidance document and practising questions using the past papers available.

Introduction to the Overall Performance of the Unit

Chemistry

SECTION B: PERIODICITY AND PROPERTIES OF ELEMENTS

Learners that were successful this year showed a good understanding of key terms and scientific vocabulary and were able to apply this understanding to given scenarios, often relaying their scientific understanding through the use of fully labelled diagrams. They were able to carry out calculations in a clear and logical manner to achieve the desired outcome.

Introduction

Physics

SECTION C: WAVES IN COMMUNICATION

The purpose of this examination is to allow learners to show their knowledge and understanding in relation to topics on 'Waves in Communication'. The topics include the study of wave features, the application of diffraction gratings and the transmission of analogue and digital signals by means of waves which are part of the electromagnetic spectrum.

Introduction to the Overall Performance of the Unit

Physics

SECTION C: WAVES IN COMMUNICATION

Learners were generally able to interpret the diagram showing a sound wave in a pipe and were able to work out the wavelength however determining the position of an antinode shown on wave diagram proved to be more difficult.

Learners have improved in their use of algebra with the majority sucessfully rearrangeing an equation. This now needs to be extended to the calculations of sin and $\sin^{-1}$ which are needed for the calculations of refractive index and critical angle as applied to various interfaces.

The importance of the critical angle in glass to give either refraction or total internal reflection was not understood by the majority of learners. However, learners were able to identify how the speed and direction of light changes when entering glass from air.

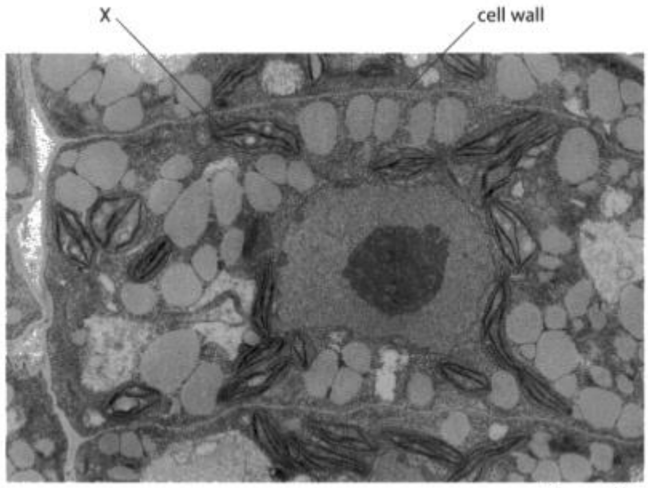
Knowledge of the appliacation of waves is improving with learners recognising how broad band is transmitted and that the bluetooth system uses radio waves.
Although in previous papers learners had been able to identify elements from emission spectra diagrams, discussing how this is done was found to be challenging and only a few learners were able to do this successfully.

Individual Questions

Biology
SECTION A: STRUCTURES AND FUNCTIONS OF CELLS AND TISSUES

Q1(a)
This question was answered well by the majority of learners. Learners should be able to recognise organelles from electron micrographs.

1 Figure 1 shows an electron micrograph of a plant cell and some surrounding cells.



(Source: ©PAL)

Figure 1

(a) Identify the cell component labelled X in Figure 1.

(1)

- ☐ **A** cell membrane
- ☒ **B** chloroplast
- ☐ **C** pit
- ☐ **D** tonoplast

Q1(b)
This question was attempted by the majority of learners. Learners tended to know that the type of ribosome was 80s, but some incorrectly stated 70s.

Lead Examiner comment: *This response was awarded 1 mark.*

(b) State the type of ribosome found in plant cells.

(1)

80S

Q1(c)

This question was answered well by the majority of learners, although many learners gave quite simplistic responses. Marks were generally lost for failing to reference absorption, learners instead referring to the 'transport' of water or nutrients without giving a direction, suggesting a confusion with the xylem or phloem. Some were even more vague, stating that the root hair cells 'help the plant grow'.

Lead Examiner comment: *This response was awarded 2 marks. We can accept 'absorb nutrients' or 'absorb ions' in place of 'absorb minerals'.*

(c) Root hair cells are specialised plant cells.

Give **two** functions of root hair cells.

(2)

1 absorb nutrients from the soil

2 absorb water from the Soil

Lead Examiner comment: *This response was awarded 0 marks. While the learner has mentioned water and minerals, they have incorrectly stated that root hair cells transport them out of the cell. Transports into the cell would be acceptable, but we do need the idea of going into the cell, not out. The vacuole comment is too generic to be creditworthy.*

(c) Root hair cells are specialised plant cells.

Give **two** functions of root hair cells.

(2)

1 transports water and minerals out of the cell.

2 large central vacuole which has cell saps, ~~and~~ regulates ~~the~~ ~~Particles~~

Q1(d)

Full marks were awarded to around half of the learners.
Most learners managing to obtain at least 2 marks for a response which was a power of ten error, due to an incorrect conversion of the observed diameter.

Lead Examiner comment: *This response was awarded 3 marks.*

(d) A plant cell is viewed using a microscope.

The observed diameter of the plant cell is 13.2 mm.

The plant cell has an actual diameter of 66 µm.

Calculate the magnification used to view the image.

(3)

Show your working.

$$\text{Magnification} = \frac{\text{image}}{\text{actual}}$$

$$\frac{13{,}200}{66{,}000} = 200$$

magnification = 200 ×

Lead Examiner comment: *This response was awarded 2 marks. The learner has not converted the original values into the same unit, but has divided the values appropriately and given a correct evaluation of that equation, therefore Error Carried Forward can be applied and 2 marks for the processes can be awarded.*

(d) A plant cell is viewed using a microscope.

The observed diameter of the plant cell is 13.2 mm.

The plant cell has an actual diameter of 66 μm.

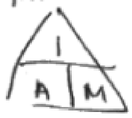
Calculate the magnification used to view the image.

(3)

Show your working.

$I = 13.2$

$A = 66\,\mu m \div 100 = \cancel{66mm}\ 0.66$



$M = \frac{I}{A}$

$M = \frac{13.2}{6.6}$ 20X

magnification = 20X

Q2(a)

Fewer than half of learners were able to answer this question correctly with a significant number not attempting it. Many of the learners who did attempt an answer showed they had knowledge of the structure of the neurone but didn't provide enough detail to gain the mark.

Among the incorrect answers, other structures of neurone were common, e.g. synaptic bulb, neurone membrane, pre-synaptic membrane, receptor(s), as were cell organelles e.g. mitochondria, nucleus, ribosome. Some learners identified Y as sodium, calcium, or just 'ions'.

Lead Examiner comment: *This response was awarded 1 mark.*

Figure 2 shows a diagram of a synapse in the brain.

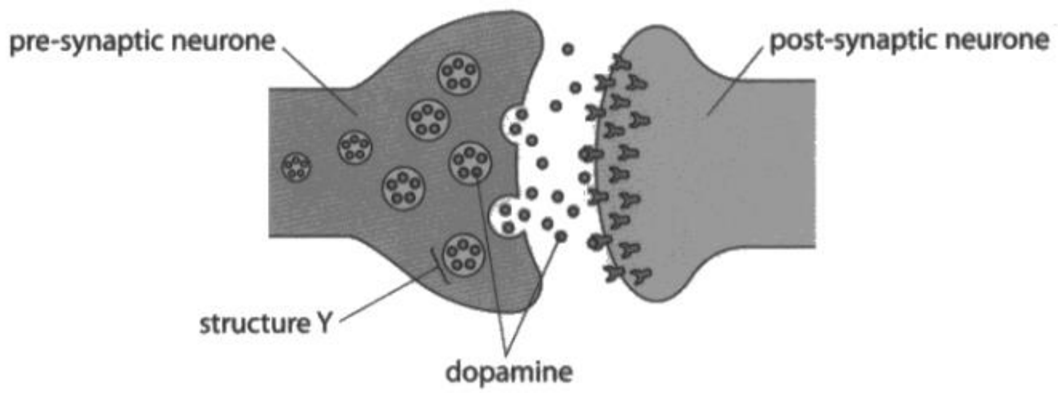


Figure 2

(a) Identify structure Y in Figure 2.

(1)

vesicle

Q2(b)

Fewer than half of learners were able to answer this question correctly to gain all 3 marks. Many of the learners who did attempt an answer showed they had knowledge of the action of dopamine at the synapse, and usually gained a mark for correctly identifying R as "neurotransmitter" or "chemical".

It was common to see 'exocytosis' or just 'transmission' for S, rather than diffusion.

Some learners were able to identify T as "receptors" or "sodium channels", and both would have been accepted, but this was often left unanswered.

Lead Examiner comment: *This response was awarded 3 marks.*

(b) Paragraph 1 partly describes the action of dopamine at the synapse.

Dopamine is a ....R............ released from the pre-synaptic membrane via exocytosis. Dopamine crosses the cleft by the process of ....S............ . Dopamine binds with ....T............ on the post-synaptic membrane.

Paragraph 1

Identify the missing words R, S and T in Paragraph 1.

(3)

R neurotransmitter

S diffusion

T receptors

Lead Examiner comment: *This response was awarded 1 mark. Neurotransmitter is an accepted response for word R and so the mark is awarded. 'Chemical' would have also been a creditworthy response for R. 'Depolarisation' and similar words connected to action potentials are common responses for S, but incorrect. Sodium without further qualified is insufficient for the mark for T. There needs to be a reference to the sodium channels or to the receptors.*

(b) Paragraph 1 partly describes the action of dopamine at the synapse.

Dopamine is a ....R.......... released from the pre-synaptic membrane via exocytosis. Dopamine crosses the cleft by the process of ....S.......... . Dopamine binds with ....T.......... on the post-synaptic membrane.

Paragraph 1

Identify the missing words R, S and T in Paragraph 1.

(3)

R ~~chemical~~ neurotransmitter

S depolarisation

T sodium

Q2(c)(i)

Lots of learners obtained a mark for Parkinson's disease or depression. Others gained marks for diseases listed by the NHS website as a disease that can be caused by an imbalance of dopamine, but are not listed on the specification. Some common incorrect answers were stroke, heart disease, cancer, Alzheimer's and COPD.

Lead Examiner comment: *This response was awarded 1 mark. 'Parkinson's' can be accepted without 'disease'.*

(c) An imbalance in dopamine levels can contribute to ill health.

(i) Name **one** disease that can be caused by an imbalance of dopamine.

(1)

Parkinsons

Q2(c)(ii)

Many correct answers gaining one mark.

Lead Examiner comment: *This response was awarded 1 mark.*

(ii) Identify the precursor molecule which can be given to raise dopamine levels.

(1

☒ **A** ATP

☒ **B** L-Dopa

☒ **C** muscarine

☒ **D** nicotine

Q3(a)

Many learners managed to achieve one mark for mentioning that the walls of the alveoli are thin. The idea that the alveoli have a good blood supply was also frequently mentioned.

Many of the learners stated more than one of the alternatives for the same marking point, which limited them to 1 mark.

Of those who failed to gain a mark, many wrote about the large surface area of the alveoli and about gaseous exchanging which were not creditworthy as this went beyond the scope of the question- the learner only had to describe the structure and not the function.

Many learners incorrectly wrote about the role of cilia or confused the alveoli for the bronchi or bronchioles.

Lead Examiner comment: *This response was awarded 2 marks. "Squamous" alone can be considered sufficient for MP2 but flattened is also MP2. The learner has also mentioned 'single layer' to achieve MP3. The learner has referenced the fourth marking point, the idea of the cells being tightly packed, which would also be creditworthy.*

3 The lungs are part of the respiratory system.

(a) Gas exchange occurs in the alveoli in the lungs.

Describe the structure of the walls of the alveoli.

(2)

Squamous cells are a single layer of flattned cells, they closely fit together like a flat stone. They are found in the alveoli of the lung. Alveoli is where oxygen are carbon dioxide exchange

Lead Examiner comment: *This response was awarded 1 mark. The learner has incorrectly explained the function and the importance of the structure. MP3 can be awarded for "thin" cell walls as this is a description of a single layer of cells. The same marking point could also be awarded for the learner's reference to a short diffusion pathway, but each marking point can only be awarded once.*

3 The lungs are part of the respiratory system.

(a) Gas exchange occurs in the alveoli in the lungs.

Describe the structure of the walls of the alveoli.

(2)

alveoli have very thin cell walls to allow diffusion to occur easier. They also have many capillaries to increase the area which oxygen can diffuse into the blood and carbon dioxide to diffuse out.

Q3(b)

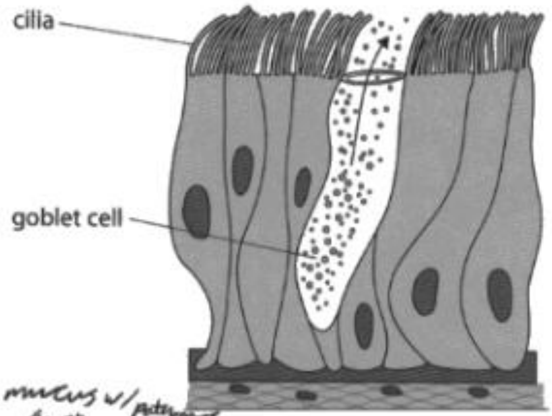
Many learners were able to explain that goblet cells secret mucus, but fewer explained that mucus trapped pathogens. Many answers included some reference to removing mucus or pathogens from the lungs.

Some learners failed to separate the functions of goblet cells and ciliated epithelium with statements such as “goblet cells and ciliated epithelium move mucus out of the lungs.” Some made reference to a “push” without further clarification.

A common misconception was that goblet cells are involved in facilitating gas exchange. Some learners confused the function of mucus and cilia. A significant number of learners confused the roles of goblet cells, mucus and cilia.

Lead Examiner comment: *This response was awarded 4 marks. The learner has clearly stated that the goblet cells produce mucus, although secrete would be preferred we can credit this as MP1. We can accept the "trap...pathogens" comment for MP2. The learner then gives two mark-worthy points for the ciliated cells in one concise phrase, as they have linked waft way the mucus (with the pathogens trapped in the mucus).*

(b) Figure 3 shows ciliated epithelial cells and goblet cells found in the lungs.



waft mucus w/ pathogens away either into stomach or out of system

Figure 3

– produce mucus

traps particles + Pathogens

Explain how the ciliated epithelial cells and goblet cells protect the lungs.

(4)

In the lungs, goblet cells produce mucus. This mucus serves 2 main purposes within the body, one is to act as a sort of lubrication for the body and the second as a defense mechanism to trap particles and pathogens, preventing illness. Once the mucus has captured pathogens, the ciliated epithelial cells 'waft away the mucus to either dispose of in stomach acid or get rid of through coughing or sneezing.

(Total for Question 3 = 6 marks)

Lead Examiner comment: *This response was awarded 1 mark. The learner has incorrectly explained the role of the goblet cells- there was a common misconception that the goblet cells catch the particles, rather than secrete mucus. The learner has not described the movement of the ciliated epithelial cells but they have explained their role and so MP4 alone can be credited.*

(b) Figure 3 shows ciliated epithelial cells and goblet cells found in the lungs.

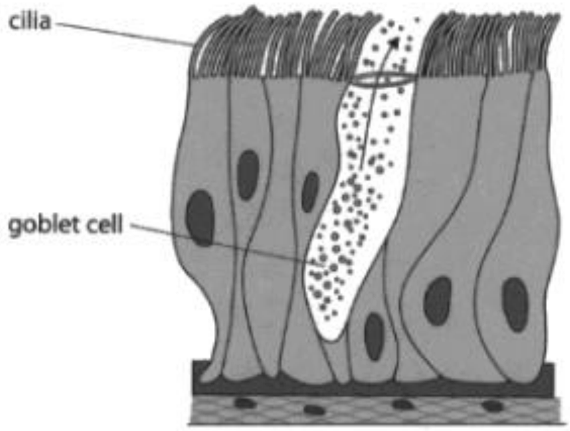


Figure 3

Explain how the ciliated epithelial cells and goblet cells protect the lungs.

(4)

~~The ciliated epithelial cells protect the lungs by using tiny hair like structures that remove bacteria and any unwanted foreign particles from the lungs~~

The goblet cell catches and ~~re move~~ removes the ~~foregn particet~~ particles away from the lungs.

The ciliated epithelial cells protect the lungs by using tiny hair like structures that remove bacteria and ~~an~~ foriegn particles from the lungs.

Q4(a)(ii)

Many learners appear to have misunderstood the question and have instead described what happens prior to the pathogen being ingested by the phagocyte.

The most common marks were given for the idea of the phagocyte breaking down the pathogen, but only a few went on to mention enzymes and even fewer mentioned the formation of a phagosome/vesicle.

Many learners gave vague descriptions of what a neutrophil is or how a neutrophil is adapted and comments about multi-lobed nuclei were commonly seen. A significant number of learners included descriptions of antibodies in their response.

Lead Examiner comment: *This response was awarded 2 marks. The learner has not achieved marks for repeating the stem, i.e., reference to pathogens being "destroyed" or "ingested", however "break down" is acceptable for digest. We do not accept the idea of "kill" for digest. The learner has also referenced enzymes for MP3.*

(ii) Neutrophils bind to pathogens. Pathogens are ingested by phagocytosis.

Describe how pathogens are destroyed by neutrophils after they are ingested.

(2)

The way in which pathogens are destroyed after being ingested by phagocytes is due to the enzyme produced by Neutrophils made to specifically break down pathogens. ~~Phag~~ Once broken down, the neutrophils can learn what Pathogen the foreign invader is.

Lead Examiner comment: *This response was awarded 0 marks. The learner has repeated the stem with ingest. "Break down" has been mentioned, but in the context of white blood cells/ antibodies locating and breaking down a pathogen and so is incorrect.*

(ii) Neutrophils bind to pathogens. Pathogens are ingested by phagocytosis.

Describe how pathogens are destroyed by neutrophils after they are ingested.

(2)

After pathogens are ingested by phagocytes, specific white blood cells; antibodies ~~attach~~ locate and break down the paltrogen.

Q4(a)(iii)

This question was answered very poorly by nearly all learners, revealing a general lack of knowledge or confusion about the role of B-lymphocytes in the immune response. Fewer than half of learners achieved any of the marking points, the most common of which was for referring to the idea of secreting antibodies. Reference to antitoxins in correct context was rarely seen.

Many learners stated that memory cells were released from B-lymphocytes, or that all B-lymphocytes were memory cells. A significant number of learners incorrectly explained that B-lymphocytes carry out phagocytosis.

Lead Examiner comment: *This response was awarded 2 marks. The learner has stated that the antibodies will be specific to antigens found on a particular pathogen- which is MP3. Antibodies binding to the antigens, with the rest of this response, is sufficient for MP1.*

(iii) Explain the role of B-lymphocytes in the immune response.

(2)

B-lymphocytes will have specific antibodies ~~antigens~~ on their cell surface membrane. These will be specific to antigens found on the cell surface membrane of a particular pathogen. These antibodies will bind to the antigens, killing the pathogen

(Total for Question 4 = 5 marks)

Lead Examiner comment: *This response was awarded 0 marks. The learner has repeated the stem and not provided any creditworthy details. The reference to "find and bind with" is not specific enough to earn a mark.*

(iii) Explain the role of B-lymphocytes in the immune response.

(2)

B-lymphocytes are a type of white blood cells that initially & find and ~~the~~ bind with the pathogen and then waits until the neutrophil ~~phagocytes~~ ingest the pathogen

Q5

This was a level based, indicative content question and was not point based. The learner's response is considered holistically and judged against the level descriptors for the targeting of the question and the command verb used.

Unfortunately, a significant number of responses were generally unstructured and consisted mainly of unconnected points which meant that the comparative element was somewhat lost. This limited the response to a maximum of 4 marks, in level 2.

Most learners were able to compare the types of cells and give some general similarities and differences, often from what was obvious from the diagram. Most common similarities were that they both contain DNA / genetic material although some learners were able to recognise that both controlled the cell.

Most common differences were that the nucleus is found in eukaryotic cells and that the nucleoid is found in prokaryotic cells and the differences in chromosome structure (linear vs circular). Many learners correctly described that the nucleus is membrane bound whereas the nucleoid is not.

Some common misconceptions were that the nucleus is surrounded by a wall, nucleoid is single stranded, and that the nucleus controls the activities of the cell while nucleoid only controlling division/movement.

Lead Examiner comment: *This response was awarded level 3, 5 marks.*
For Level 3 (5 or 6 marks) we are looking for a comparison between the nucleus and nucleoid- here the comparison is obvious throughout. This might not always be the case but the command verb "compare" should be fulfilled for a mark in band 3. The structure should be logical and clear. Here the learner has logical chains of reasoning supported by application of knowledge.

The learner does not have to give all the details from the indicative content but there does have to be sufficient knowledge demonstrated to be "comprehensive". This example is just sufficient to be classed as more than good knowledge and so just fulfils the requirements to be classed as comprehensive, although there is an error in their knowledge (which slightly affects the mark awarded).

The learner has given correct information about the nucleus with reference to histones and being condensed into chromosomes. They have provided a similarity (both contain DNA) and several differences have been mentioned- we would expect to see this for a band 3 response- the response should be more *balanced with similarities and differences. The function on page two of the response is incorrect about prokaryotes and mitosis and meiosis.*

This response overall is a band 3 response and has been awarded 5 marks.

Compare the structures and functions of a nucleus and a nucleoid.

Your answer should refer to similarities and differences.

(6)

A nucleus and a nucleoid both contain DNA. However, a nucleus has a membrane whereas a nucleoid does not. Also, the DNA contained in a nucleus is condensed into chromosomes, whereas in nucleoids it is not. Furthermore, the DNA found in a nucleoid is short and doesn't contain histones whereas the DNA in a nucleus is longer and does contain histones. A nucleus is found in eukaryotic cells whereas a nucleoid is found in prokaryotic cells. The function of both the nucleus and the nucleoid is to store DNA and replicate DNA during cell division (mitosis and meiosis)

Lead Examiner comment: *This response was awarded level 2, 4 marks. This response is in the middle of level 2- it is a borderline response between 3 and 4 marks. The learner has given a couple of similarities about the DNA and controlling the cell. They have given some information about the nucleus and its shape, and about RNA, and so there is a suggestion of some differences and so a suggestion of comparison. Overall, this is a level 2 response and just creditworthy for 4 marks.*

Compare the structures and functions of a nucleus and a nucleoid.

Your answer should refer to similarities and differences.

(6)

Similarities - Both contain DNA
- Both the nucleus and nucleoid control the cell

Differences - Nucleus is one circle whereas nucleoid is ~~a sigle~~ single circular strand of DNA
~~Whereas~~ figure 5 a is ~~pro karyo~~ eukaryotic as it has a nucleus prokaryotic in figure 5 b as it has a nucleoid. N~~a~~ucleus makes RNA. ~~So~~ Nucleus surrounded by many things whereas Nucleoid is surrounded by less ~~things~~. organelles.

Lead Examiner comment: *This response was awarded level 1, 2 marks. This is an example of a level 1 response. The learner has given a lot of irrelevant information about generic cell organelles.*

Here the learner has stated that they both have genetic material- which is a similarity- and they have mentioned the nucleolus-which is a difference.

They have also mentioned a function- controls the cells- but this is a little vague. Overall, this response can be awarded 2 marks.

Compare the structures and functions of a nucleus and a nucleoid.

Your answer should refer to similarities and differences.

(6)

Both figure ~~a~~ 5a (animal cell) and figure 5b (bacteria) have ribosomes to produce ~~prod~~ protein. They both have genetic material, however this is stored in the nucleus of 5a rather than being free in the cell of 5b. 5b has a flagellum to move around and pili to attach to things. It also has a capsule and a cell wall. 5a has none of these as it stays in place. The genetic material in 5b is stored in the nucleoid as it is more free to move around, which is ~~needed~~ needed when 5b moves around alot.

The nucleus is ~~spheroid~~ ~~circular~~ circular from a 2D perspective and has a nucleolus inside it. It controls the actions of the cell.

Summary

Biology

SECTION A: STRUCTURES AND FUNCTIONS OF CELLS AND TISSUES

Based on their performance on this paper, learners should:

- Understand the demand of the command verbs, especially the difference between describe and explain.
- Be familiar with and recognise the ultrastructure and function of organelles in the following cells: prokaryote cells (bacterial cells), eukaryotic cells (plant and animal cells) and eukaryotic cells (plant-cell specific).
- Revise key definitions with Level 3 detail.
- Show their working out for all calculations to ensure marks can be given for the stages within the calculation even if the final answer is incorrect.
- Use the additional guidance to understand the depth and breadth of the specification. The link to the Specification and/or Sample Assessment Materials (SAMs) is located on the BTEC Nationals qualification webpage here.

Individual Questions

Chemistry
SECTION B: PERIODICITY AND PROPERTIES OF ELEMENTS

Q1

The first question on the paper, **1(a)(i),** was a multiple-choice question which asked learners to identify the position of a d-block element in the periodic table. The majority of learners performed well in this question.

Part (ii) of question 1(a) was less well answered, with the majority not scoring.
The majority of learners did not read the question carefully and gave a physical property such as high melting point or the ability to conduct electricity rather than a chemical property as in these examples which both scored no marks.

(ii) State **one** chemical property of a d-block element.
(1)

High melting point

(ii) State **one** chemical property of a d-block element.
(1)

They conduct electricity.

Some simply named an element in the d-block or stated that they were metallic, again this gained no credit.

(ii) State **one** chemical property of a d-block element.
(1)

metalic

(ii) State **one** chemical property of a d-block element.
(1)

Manganese

Of those that did score, they often did so for stating that the metals have multiple oxidation states or that they are catalysts which was allowed. The following two examples scored 1 mark.

(ii) State **one** chemical property of a d-block element.

(1)

can have more than one oxidation state

(ii) State **one** chemical property of a d-block element.

(1)

They are good catalyst

Part (b)(i) of question 1 asked learners to explain why the relative atomic mass of platinum is not a whole number. A good proportion of learners were able to access the question and were able to score 2 or 3 marks. Many knew that platinum exists as isotopes and then tried to give a definition of an isotope, this example scored 1 mark.

(b) Platinum is a d-block element.

(i) The relative atomic mass of platinum is 195.1.

Explain why the relative atomic mass of platinum is not a whole number.

(3)

Platinum is a transition element, ~~the Have different no~~ It is also a Isotope (same number of atom ~~be~~ but different protones and electrons.

Some learners showed how they would carry out a calculation of an isotope, this showed a good understanding and by showing this, it fulfilled all marking points and scored the full 3 marks.

(b) Platinum is a d-block element.

(i) The relative atomic mass of platinum is 195.1.

Explain why the relative atomic mass of platinum is not a whole number.

(3)

Because the relative atomic mass is the average of the mass of all the isotopes of platinum, weighted by how abundant they are compared to the other isotopes, so to calculate relative atomic mass you would do the equation: $\frac{\text{(mass of isotope x relative abundance) + (mass of isotope x relative abundance)}}{100}$ which would not normally leave you with a whole ~~number~~ number.

Of those that did not score, a common misconception was that the atomic mass was a decimal number due to the number of electrons present in the atom or that the atom did not have a whole number of neutrons. Others thought that it was decimal as this was the accurate value but could take this no further. This example scored no marks.

(b) Platinum is a d-block element.

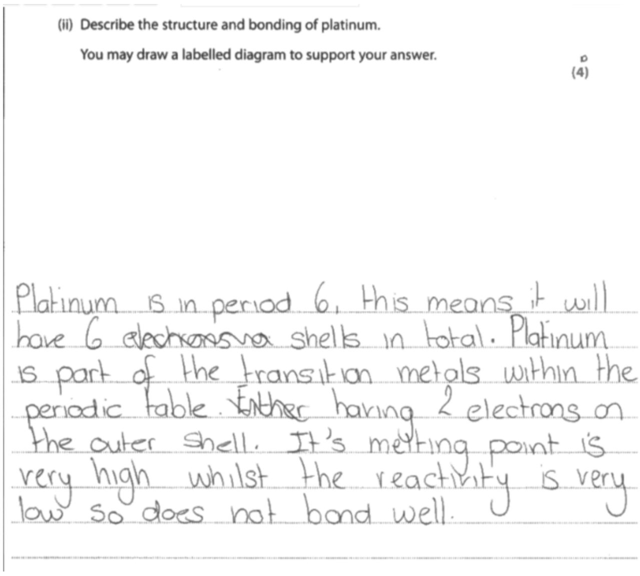
(i) The relative atomic mass of platinum is 195.1.

Explain why the relative atomic mass of platinum is not a whole number.

(3)

Because the mass of the neutrons are not a whole number.

In part (ii) learners were asked to describe the structure and bonding of platinum. Many learners did not read the question carefully and attempted to describe the electronic or the atomic structure, with many trying to draw the electronic configuration and often trying to explain a property of platinum such as high melting point or its ability to conduct electricity. These examples scored no marks.

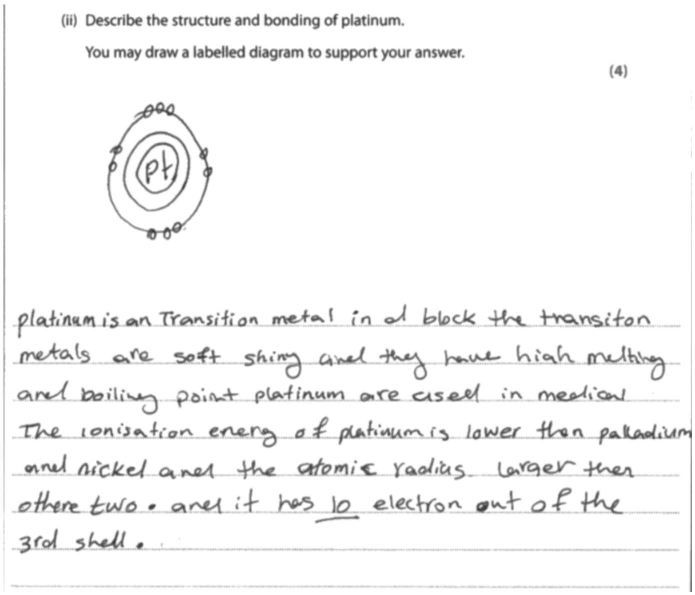


(ii) Describe the structure and bonding of platinum.

You may draw a labelled diagram to support your answer.

(4)

Platinum is in period 6, this means it will have 6 ~~electrons~~ shells in total. Platinum is part of the transition metals within the periodic table. Either having 2 electrons on the outer shell. It's melting point is very high whilst the reactivity is very low so does not bond well.



(ii) Describe the structure and bonding of platinum.

You may draw a labelled diagram to support your answer.

(4)

platinum is an Transition metal in d block the transiton metals are soft shiny and they have high melting and boiling point platinum are used in medical The ionisation energy of platinum is lower than palladium and nickel and the atomic radius larger then othere two. and it has 10 electron out of the 3rd shell.

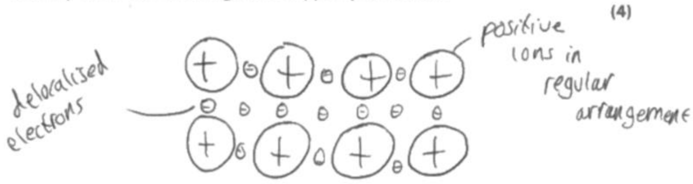
Of those that did read the question carefully, a good proportion scored with many drawing the structure of the metal, showing the regular lattice of metal cations and delocalised electrons. Most were then able to state that the metal has metallic bonding, fewer were then able to describe the metallic bond as the electrostatic attraction between the metal cations and the delocalised electrons. Many confused types of bonding and often contradicted references to metallic bonding with references to ionic, covalent or intermolecular bonding.

This example shows a good response that scored the full 4 marks available, with two of the marks being scored from the labelled diagram.

(ii) **Describe the structure and bonding of platinum.**

You may draw a labelled diagram to support your answer.

(4)



Platinum is a metallic compound meaning its structure consists of a sea of delocalised electrons within a giant lattice of positive ions. There is strong electrostatic forces of attraction between the negative electrons and positive ions, which requires a lot of heat energy to be overcome. When two metals react they will form a strong metallic bond.

Q2

Question 2 focused on the alkali metals. **Part (a)** asked learners to state, in terms of electronic configuration, why the elements are all placed in the same group. A good proportion of learners were able to answer this concisely and understood that they were put in the same group because they had the same number of electrons in the outer shell, some went further and stated that they all have one electron in their outer shell both of which scored.

Both of these examples scored the 1 mark available.

2 Lithium, sodium and potassium are some of the elements in group 1 of the periodic table.

(a) State, in terms of their electronic configuration, why these elements are all placed in the same group.

(1)

All these elements ~~are in the~~ have the Same amant of electrons on there outer Shell.

2 Lithium, sodium and potassium are some of the elements in group 1 of the periodic table.

(a) State, in terms of their electronic configuration, why these elements are all placed in the same group.

(1)

They all have 1 electron in their outer shell because they are all in group 1.

Some learners were not precise with their answers and just stated that the elements have the same number of electrons, with no reference to the outer shell electrons, as in this example which scored no marks.

2 Lithium, sodium and potassium are some of the elements in group 1 of the periodic table.

(a) State, in terms of their electronic configuration, why these elements are all placed in the same group.

(1)

they have the same number of electrons. they can carry the same number of electrons

Where learners did not score, it was often because they did not read the question carefully and answered in terms of reactivity rather than electron configuration stating that they all react in the same way, others simply stated that they all had electrons in the s orbital which was insufficient for credit. This answer scored no marks.

2 Lithium, sodium and potassium are some of the elements in group 1 of the periodic table.

(a) State, in terms of their electronic configuration, why these elements are all placed in the same group.

(1)

They are all reactive metals.

Part (b) of question 2, asked learners to consider the reaction of potassium in water. In **part (i)**, many were able to identify the colour flame produced by potassium when it is reacted in water, as lilac correctly and a good proportion were able to add the state symbols to the balanced equation for the reaction in **part (ii)** to score 2 marks as in this example.

(ii) The balanced equation for the reaction is

$$2K(s) + 2H_2O(l) \rightarrow 2KOH(aq) + H_2(g)$$

Add the state symbols **aq**, **g**, **l** and **s** to the balanced equation.

You must only use each state symbol **once**.

(2)

A common error was to think that water was aqueous and KOH was a liquid, but most scored at least 1 mark for (s) and (g) or any two correct state symbols in the correct place as in this example.

(ii) The balanced equation for the reaction is

$$2K(s) + 2H_2O(aq) \rightarrow 2KOH(l) + H_2(g)$$

Add the state symbols **aq**, **g**, **l** and **s** to the balanced equation.

You must only use each state symbol **once**.

(2)

Learners found **part (iii)** more difficult. Many learners were not clear as to which element was reduced in the reaction. Many thought that it was potassium, oxygen or water that was reduced. Of those that did understand that hydrogen was reduced, few were able to explain this in terms of oxidation number as requested in the question and either mentioned loss or gain of electrons or oxygen which did not score.

Some learners referred to hydrogen gas being reduced rather hydrogen (in the water) and so did not score the mark.

(iii) Explain, in terms of change in oxidation number, which element is **reduced** in the reaction.

(2)

Hydrogen gas was reduced

In this example the learner has shown a good understanding and explained in terms of change in oxidation number to gain both marks.

(iii) Explain, in terms of change in oxidation number, which element is **reduced** in the reaction.

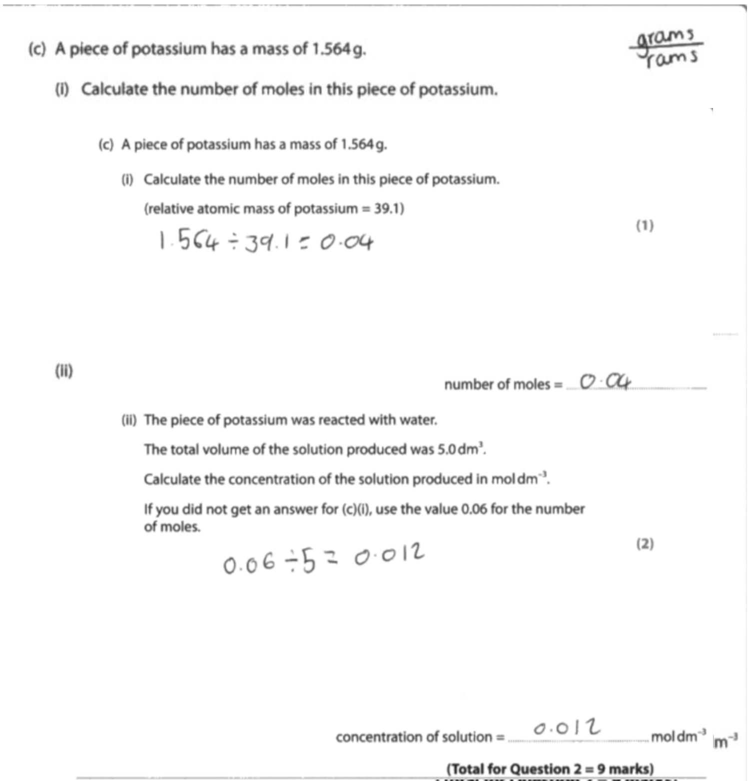
(2)

Hydrogen is the reduced atom since it is seperated it ~~an~~ now has 0 charge but as water it was plus 1 so it has gained an electron

Part (c) (i) was the first calculation in the paper, where learners were asked to calculate the number of moles of potassium in a piece of mass of 1.564g. It was pleasing to see that the majority of learners were able to carry out the calculation correctly to score the mark.

Calculating the concentration using the number of moles from **part (i)** proved more difficult for learners but still with a good proportion scoring both marks for dividing their number of moles calculated by the volume given and then evaluating this as 0.008 or $8x10^{-3}$.

This example scored 1 mark on **part (i)** and then 2 marks on **part (ii)**



(c) A piece of potassium has a mass of 1.564 g.

grams
rams

(i) Calculate the number of moles in this piece of potassium.

(c) A piece of potassium has a mass of 1.564 g.

(i) Calculate the number of moles in this piece of potassium.

(relative atomic mass of potassium = 39.1)

(1)

1.564 ÷ 39.1 = 0.04

(ii)

number of moles = 0.04

(ii) The piece of potassium was reacted with water.

The total volume of the solution produced was 5.0 dm^3.

Calculate the concentration of the solution produced in mol dm^{-3}.

If you did not get an answer for (c)(i), use the value 0.06 for the number of moles.

(2)

0.06 ÷ 5 = 0.012

concentration of solution = 0.012 mol dm^{-3} m^{-3}

(Total for Question 2 = 9 marks)

In some cases, where learners did not get an answer for **part (i)** or were unsure of their answer, they used the value of 0.06 for the number of moles giving an answer of 0.012 or $1.2x10^{-2}$. This example scored 1 mark on **part (i)** and 2 marks on **part (ii).**

In this case, where the learner had made an error in **part (i)**, this error was carried forward and so even though the learner did not score in **part (i)** they were still able to gain full marks in part (ii). This learner scored 0 marks on **part (i)** and 2 marks on **part (ii).**

(c) A piece of potassium has a mass of 1.564 g.

(i) Calculate the number of moles in this piece of potassium.

(relative atomic mass of potassium = 39.1)

(1)

$1.564 \times 39.1 = 61.1524$

number of moles = 61.1524

(ii) The piece of potassium was reacted with water.

The total volume of the solution produced was 5.0 dm^3.

Calculate the concentration of the solution produced in mol dm^{-3}.

If you did not get an answer for (c)(i), use the value 0.06 for the number of moles.

(2)

$\frac{0.06}{5} = 0.012$

$\frac{61.1524}{5} = 12.23048$

concentration of solution = 12.23048 mol dm^{-3}

(Total for Question 2 = 9 marks)

Q3

Question 3 focused on a model of an atom. In **part (a)(i)** learners were asked to complete the first and third subshells, a good proportion scored both marks as in this example.

3 (a) Figure 2 shows a model of an atom.

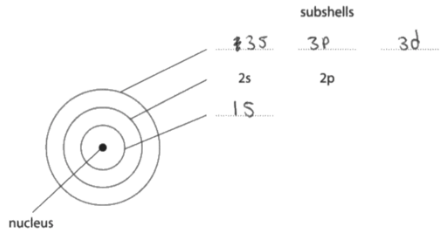


Figure 2

(i) The model shows three shells of electrons.

These shells can be split into subshells.

The subshells of the second shell have been completed for you.

Complete Figure 2 to show the subshells in the first and third shells.

(2)

In some cases, learners confused the 3d with the 3f subshell and so scored just 1 mark as in this example.

3 (a) Figure 2 shows a model of an atom.

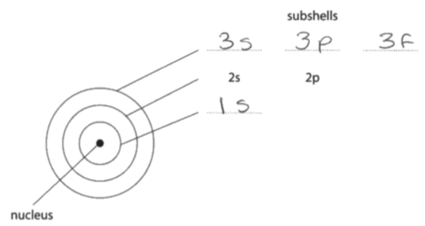


Figure 2

(i) The model shows three shells of electrons.

These shells can be split into subshells.

The subshells of the second shell have been completed for you.

Complete Figure 2 to show the subshells in the first and third shells.

(2)

In **part (ii)**, the vast majority of learners knew that the 2p subshell contained 6 marks to gain the mark as in this example.

(ii) State the maximum number of electrons that can occupy the 2p subshell.

(1)

6

Where learners lost marks it was often as they gave an answer of 2 or 8, possibly thinking of shells rather than sub shells, as in these examples which both scored no marks.

(ii) State the maximum number of electrons that can occupy the 2p subshell.

(1)

[illegible] 2

(ii) State the maximum number of electrons that can occupy the 2p subshell.

(1)

8

Learners found **part (b)** much more difficult with few scoring the full three marks available. Where learners did score, it was usually for stating that for the 2nd electron affinity oxygen was negative, but of these few were then able to explain that this would cause a repulsion with the electron being gained.

A common misconception was that initially oxygen was positive and others confused electron affinity with ionization energy and talked about electrons being removed.

(b) Electron affinity is the heat energy change when one mole of gaseous atoms gains one mole of electrons.

Table 1 shows the equations and the energy change for the first two electron affinities of oxygen.

	equation	energy change
first electron affinity	$O(g) + e^- \rightarrow O^-(g)$	releases energy
second electron affinity	$O^-(g) + e^- \rightarrow O^{2-}(g)$	requires energy

Table 1

Explain why the first electron affinity of oxygen releases energy but the second electron affinity of oxygen requires energy.

(3)

- It needs energy to get rid of the electronsin the second electron affinity.

Q4

Question 4 was well answered and differentiated well between learners. A good proportion of learners were able to draw a correct dot and cross diagram for chlorine and discuss how the covalent bond formed. Many did mention intermolecular forces and quite a few named the correct intermolecular force. Those who went on to also describe how induced dipoles occurred often achieved level 3. A few however seemed to think that the intermolecular force was permanent dipole-dipole, even though they sometimes mentioned that chlorine was non-polar, which showed a lack of understanding. Those who included diagrams in their answer often scored higher marks than those who didn't. Most gained at least 1 mark by mention of covalent bonding.

In this example, the learner has given a correct dot and cross diagram with nothing further this gained full credit at Level 1 with 2 marks.

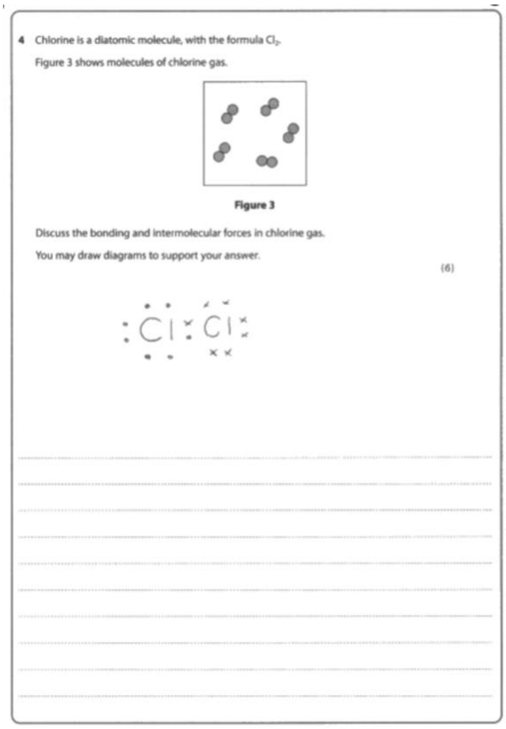

4 Chlorine is a diatomic molecule, with the formula Cl_2.

Figure 3 shows molecules of chlorine gas.

Figure 3

Discuss the bonding and intermolecular forces in chlorine gas.

You may draw diagrams to support your answer.

(6)

In some cases, learners did not address the covalent bonding in the molecule but only the intermolecular forces. In this example, the learner has given a very good account of how the London dispersion forces are formed, which gained full credit at Level 2 with 4 marks.

4 Chlorine is a diatomic molecule, with the formula Cl_2.

Figure 3 shows molecules of chlorine gas. Cl = Cl

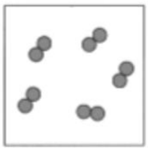


Figure 3

Discuss the bonding and intermolecular forces in chlorine gas.

You may draw diagrams to support your answer.

(6)

Cl_2 experience london dispersion force.
london dispersion force is caused by random electron movement. this means all atom or molecule experience london dispersion even if they experience other intermolecular forces.
Temporary uneven distribution in one molecule/atom induces dipole in the neighbouring molecule - causing a weak electrostatic force of attraction between the molecules.
- weak and easily broken - this means have low melting/boiling points - don't require a lot of energy.

(Total for Question 4 = 6 marks)

TOTAL FOR PAPER = 30 MARKS

In this last example, the learner starts with a correct dot and cross diagram labelled covalent bond. In the diagram they also show that the molecule is non-polar and on the second page, the learner goes on to state that chlorine has temporary dipole induced dipole forces and discuss how the temporary dipoles are formed with good detail to score the full 6 marks at Level 3.

4 Chlorine is a diatomic molecule, with the formula Cl_2.

Figure 3 shows molecules of chlorine gas.

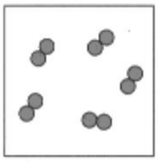


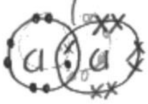
Figure 3

Discuss the bonding and intermolecular forces in chlorine gas.

You may draw diagrams to support your answer.

(6)

covalent bond



each chlorine atom has 7 outer electrons

Cl — Cl

same ~~intermolecular~~ electronegativity = no polar bonds

symmetry = non-polar molecular

Chlorine is a diatomic molecule meaning each molecule of Chlorine consists of 2 chlorine atoms. Each chlorine atom has 7 outer shell electrons meaning 2 chlorine atoms can form a covalent bond by sharing a pair of electrons (one from each atom).

In addition, because the 2 chlorine atoms have the same electronegativity and when they form a covalent bond it is symmetrical this means the ~~mo~~ chlorine molecule is non-polar.

This means chlorine has the weakest intermolecular forces of attraction which is the temporary dipole - induced dipole forces which is also the weakest out of the Van de Waals Forces. This is where random electron movement in one molecule creates a temporary ~~dip~~ dipole which then induces a dipole in another molecule. This means chlorine has a relatively low melting/boiling point as ~~not much~~ little heat energy is required to overcome the weak intermolecular forces.

(Total for Question 4 = 6 marks)

TOTAL FOR PAPER = 30 MARKS

Summary

Chemistry
SECTION B: PERIODICITY AND PROPERTIES OF ELEMENTS

Based on their performance on this paper, learners should:

- ensure they read the questions carefully, consider highlighting key terms and/or command words to enable them to focus on the question in hand.
- re-read their answers when they have finished to check that the answer, they have given answers the question set rather than a question that they think might be there.
- use past papers to practice exam technique.
- practice laying out calculations and showing their working in a clear and logical way.
- use the additional guidance document to ensure that all subject content has been covered.
- avoid putting everything they know about a subject in an answer, try to be concise and only give detail that is relevant to the question posed.

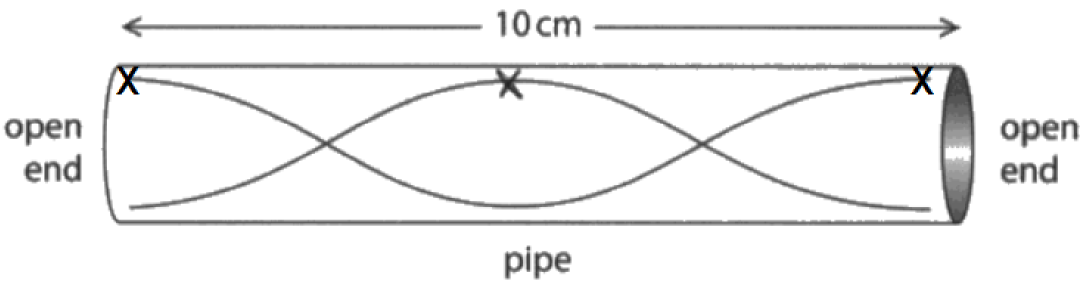
Individual Questions

Physics

SECTION C: WAVES IN COMMUNICATION

Q1

The diagram below relates to **1(a)(i)**, **1(a)(ii)** and **1(c)**



1(a)(i) More than half of learners were able to select 10cm as the wavelength of the stationary wave shown in the diagram.

1(a)(ii) More than half of learners were able to mark with an X an antinode on the diagram of a stationary wave produced in an open-ended pipe. The diagram shows three possible positions of antinodes and an X placed vertically below any X shown on the diagram would gain the mark.

1(b) The majority of learners showed their working, were able to substitute into the given equation and rearrange the equation to evaluate the wavelength of a wave as in the example below.

(b) The frequency (f) of a sound wave is 1370 Hz.

The velocity (v) of the sound is 330 ms^{-1}

Calculate the wavelength (λ) of the wave.

Use the equation $v = f \times \lambda$.

Show your working.

(3)

$330 = 1370 \times \lambda$

$\lambda = \frac{330}{1370}$

$= 0.24$

wavelength (λ) = 0.24 m

The most common error was in the incorrect rearrangement giving an answer of 4.15. However showing the working allows the example shown below to be awarded 2 marks

(b) The frequency (f) of a sound wave is 1370 Hz.

The velocity (v) of the sound is 330 ms^{-1}

Calculate the wavelength (λ) of the wave.

Use the equation $v = f \times \lambda$.

Show your working.

(3)

$330 = 1370 \times ?$

$\frac{1370}{330} = 4.15$

wavelength (λ) = 4.15 m

1(c) Learners found this part of question 1 the most challenging. Many learners were able to suggest lengthening or shortening the pipe but did not refer to the information in the diagram and suggest closing the pipe at one end. The other possible answer was to change the way the air was blown through the pipe, blow harder or softer was sufficient. The diameter of the pipe was irrelevant as all the pipes shown for the instrument have the same diameter.

(c) Give **two** ways that the frequency of the sound wave in Figure 2 can be changed.

(2)

1 Closing the pipe end

2 extending the legnth

The example below was not awarded any marks.

(c) Give **two** ways that the frequency of the sound wave in Figure 2 can be changed.

(2)

1 ~~length~~ increasing or decreasing wavelength.

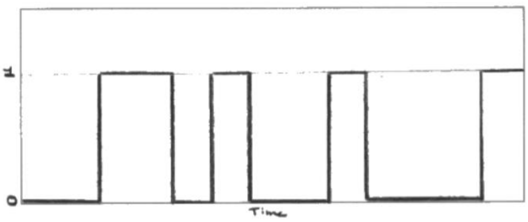
2 increasing or decreasing the velocity

Changing the wavelength was not awarded a mark as it is changes to the instrument that are required. If it had been the velocity of <u>air</u> that was to be changed then a mark would have been awarded as this is equivalent to blowing harder.

Q2

2(a) About half of learners could draw the shape of a digital signal and provided the drawing did not have sloping lines the mark was awarded.

Examples of acceptable drawings are shown below. It is best if a ruler is used to draw the diagram although a good attempt at straight lines drawn freehand was not penalised. However, it should be noted that the instructions on the front of the examination paper state that learners must have a calculator and a ruler.



2(b)(i) Learners were told that coaxial cables are used to send and receive broadband signals and asked to give two other methods of transmission. The majority of learners were able to gain one mark, giving optical fibres or satellite as a means of transmission of broadband, radio-waves and microwaves were also mentioned. However, a large number of learners repeated the stem of the question by stating that cables were used as shown in the example below

(b) Broadband is a system that gives rapid internet access.

Coaxial cables can be used to send and receive broadband signals.

(i) Give **two other** methods that can be used to send and receive broadband signals.

(2)

1 Optical Fibres

2 Copper cables/wires

2(b)(ii) When many signals are transmitted at the same time the process is called multiplexing. The majority of learners were unable to recall this answer.

Q3

3(a) This multiple choice question was found to be difficult because two facts had to be recalled. Many learners knew that light travelled slower in air than in glass however less than half of learners could identify this with the correct change in direction of the light ray as it enters the glass.

3(b) This question was a calculation that required learners to 'show that' the critical angle in glass for a ray of light passing from glass of refractive index 1.52 into air was approximately 41°. The majority of learners were able to achieve at least 1 mark either by correctly substituting into the given equation or by completing a correct rearrangement. No marks were awarded if 41° was inserted into the equation, this value had to be shown, not used.

See the example below

(b) Figure 5 shows a ray of light at a glass–air interface.

The ray of light strikes the surface of the glass at the critical angle *c*.

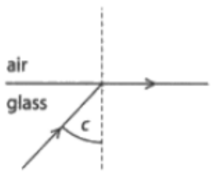


Figure 5

The refractive index (*n*) of the glass is 1.52

Show that the critical angle *c* is approximately 41°

(4)

Use the equation

$$n = \frac{1}{\sin c}$$

$$n = \frac{1}{\sin(41)}$$

$$1.52 = \frac{1}{\sin(41)}$$

critical angle *c* ....41........ °

If learners were familiar with the trigonometric functions on their calculators then this calculation did not present a problem. Learners need to be able to use these functions for calculations using critical angles or the refractive index equation.

The example below is awarded 3 marks. The substitution and rearrangement are correct giving the first two marks. Sin c has then been found for the third mark but the value of sin c has been rounded to two decimal places, 0.66 giving an incorrect value for the angle c. The value of the sin of an angle must be taken to at least 3 decimal places to give accurate values of the angle. As a rule values should not be rounded within a calculation, leave any rounding to the final answer.

(b) Figure 5 shows a ray of light at a glass–air interface.

The ray of light strikes the surface of the glass at the critical angle *c*.

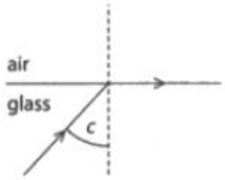


Figure 5

The refractive index (*n*) of the glass is 1.52

Show that the critical angle *c* is approximately 41°

(4)

Use the equation

$$n = \frac{1}{\sin c}$$

$1.52 = \frac{1}{\sin c}$ $\sin^{-1}(0.66) = 41.3 = 41$

$\sin c \times 1.52 = 1$

$\sin c = \frac{1}{1.52}$

$\sin c = 0.66$

critical angle *c* ....41.3........ °

This next example was awarded the full 4 marks. The answer is correct because the vaue obtained for sin c has been used in $\sin^{-1}$ c to calculate the value of the critical angle accurately.

(b) Figure 5 shows a ray of light at a glass–air interface.

The ray of light strikes the surface of the glass at the critical angle *c*.

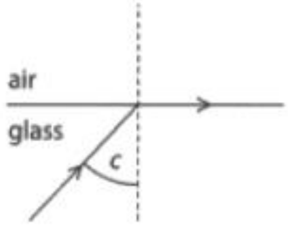


Figure 5

The refractive index (*n*) of the glass is 1.52

Show that the critical angle *c* is approximately 41°

(4)

Use the equation

$$n = \frac{1}{\sin c}$$

$n = \frac{1}{\sin c}$

$1.52 = \frac{1}{\sin c}$

$\sin c = \frac{1}{1.52}$

$\sin c = 0.6578947368$

$c = \sin^{-1}(\ \wedge\ )$

$c = 41.1395104\,1$

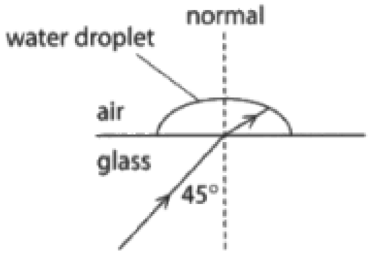
$= 41.1$

critical angle *c* 41.1 °

3(c) Learners found the analysis of the information given in the question challenging as it was necessary to link the diagrams to the critical angles given and to understand the relevance of critical angle to refraction and total internal reflection. The response below gives the complete answer and gains 3 marks. The learner has used all the information and added it to the diagrams.

(c) Figure 6a shows light incident at 45° at a glass–water interface.

Figure 6b shows light incident at 45° at a glass–air interface.



c= 62 **Figure 6a**

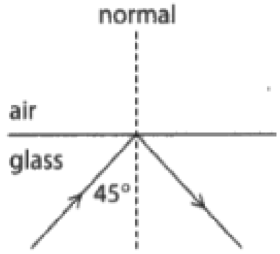


Figure 6b c=41

The critical angle for a glass–water interface is 62°
The critical angle for a glass–air interface is 41°

Explain why light is **refracted** in Figure 6a but light is **reflected** in Figure 6b.

(3)

In 6a, the angle of incidence is 45°, which is less than the critical angle ~~x~~ 62°, so the light is ~~f~~ refracted. In 6b, the angle of incidence is 45°, whichis greater than the critical angle of 41°. All of the light is totally internally reflected in 6b.

The next response gains two marks.

The learner shows an understanding of the relevance of the critical angle but has not related this to the angles given in the question

(c) Figure 6a shows light incident at 45° at a glass–water interface.

Figure 6b shows light incident at 45° at a glass–air interface.

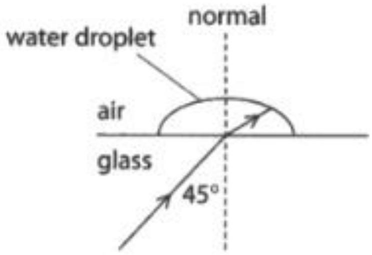


Figure 6a

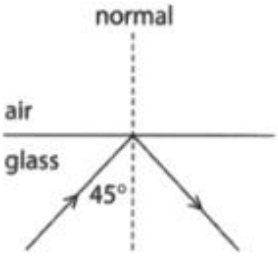


Figure 6b

The critical angle for a glass–water interface is 62°
The critical angle for a glass–air interface is 41°

Explain why light is **refracted** in Figure 6a but light is **reflected** in Figure 6b.

(3)

light refracted in figure 6a because the critical angle is greater then the incident angle where is in figure 6b it reflected because ~~on~~ the critical angle is less then the incident angle

Q4

4(a) More than half of learners could select the correct response knowing that electromagnetic waves are transverse waves and that electromagnetic waves have different frequencies.

4(b) More than half the learners were able to gain one mark in response to this question. The learners realised that in large cities there are many more mobile phone users, this was an expansion mark. However this was rarely linked to the limited capacity of the base station receiving the mobile phone signals. An example of a response gaining both marks is given below

(b) Mobile phone networks use tower masts as base stations.

Base stations receive signals from mobile phones.

Explain why a large city needs to have many base stations.

(2)

A LARGE CITY HAS A HIGH POPULATION, ALL CARRYING PHONES. THIS INCREASES THE AMOUNT OF SIGNALS SENT. A BASE STATION CAN ONLY HANDLE A LIMITED NUMBER OF SIGNALS, SO IN ORDER TO PROCESS THEM ALL IT'S NECESSARY TO HAVE MULTIPLE STATIONS.

Another possible explanation was that the signal gets weaker at a greater distance from the base station so more base stations are needed for area coverage.

(b) Mobile phone networks use tower masts as base stations.

Base stations receive signals from mobile phones.

Explain why a large city needs to have many base stations.

(2)

because the ~~connection~~ signals get weaker the further away you are so more towers are ~~bild~~ build so you're close to one always meaning the signal is stronger

4(c) Learners found this question difficult as many could not identify that Bluetooth© operated in the radio-wave region of the electromagnetic spectrum and therefore 'line of sight' was not necessary. A property of radio-waves is the ability to diffract around obstacles or pass through walls, this was then needed for the expansion of the explanation.

(c) Bluetooth® is a system that is used to connect electronic devices.

Explain why Bluetooth® signals do not need 'line of sight' to communicate with other Bluetooth® devices.

(2)

Because bluetooth uses radio waves which can travel/penetrate through walls so that they can connect with other bluetooth devices without being in a direct line of site.

Q5

This is a levelled question and the majority of learners were able to gain some marks, usually for discussing the identification of elements in the gas. The diagram for this question showed the emission spectrum of a gas and then required learners to discuss how emission spectra are produced and how the emission spectrum of the gas can be used to identify the elements present in a gas.

This is an example of a Level 1 response. There is a correct statement ‘Each element in a gas has a unique pattern of coloured lines’ but no further discussion.

5 Figure 7 shows an example of an emission spectrum of a gas.

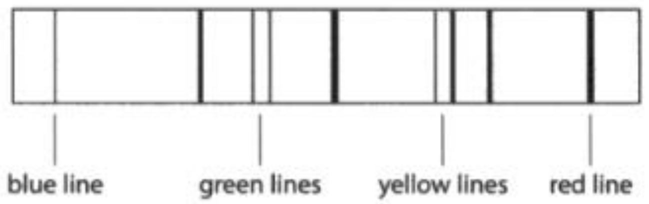


Figure 7

Discuss how emission spectra are produced and can be used to identify the elements in a gas.

(6)

~~Elements in a gas have specy a~~ .
Each element in a gas has a unique pattern of coloured lines on the emission spectrum ~~which make~~ ~~to you~~ which makes ~~the ete~~ it easier to identify what elements are in the gas.

The following responses are Level 2**.**

This response attempts an incomplete explanation of how emission spectra are produced and refers to different coloured lines for each element.

5 Figure 7 shows an example of an emission spectrum of a gas.

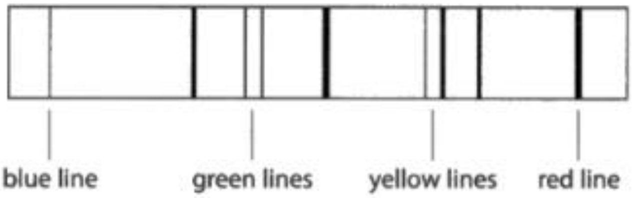


Figure 7

Discuss how emission spectra are produced and can be used to identify the elements in a gas.

(6)

Emission spectras are produced using a heat lamp to heat the gas. The electrons absorb the energy when heated, causing them to move up energy levels. This produces different coloured lines for each element as they will each have a different amant of electrons and shells, allowing you to identify the different elements within the gas.

The following example has a very good description of how the emission spectrum is produced but does not discuss correctly how the elements can be identified from the spectrum of the gas

5 Figure 7 shows an example of an emission spectrum of a gas.

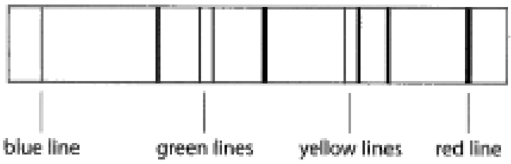
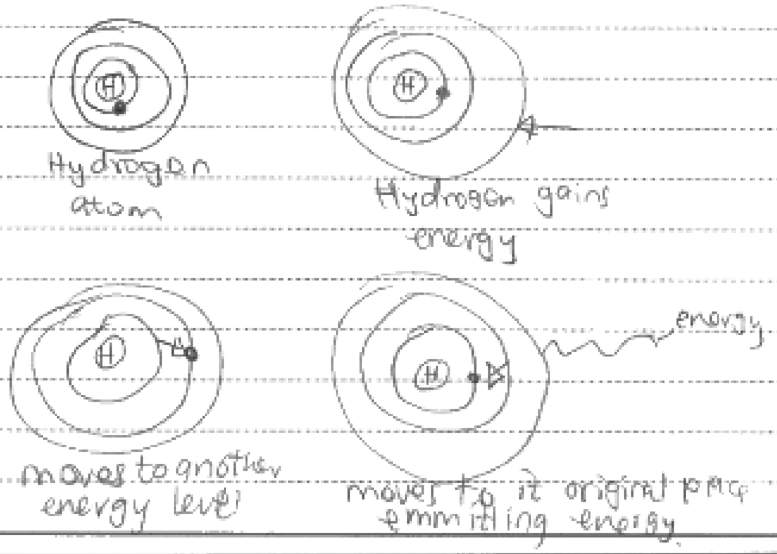


Figure 7

Discuss how emission spectra are produced and can be used to identify the elements in a gas.

(6)

An electron is usually in ground state. The electron is heated, gains energy move from it energy level to another. The electron is said to be excited. After sometime the electron move back to its energy level and emmits the energy produced a spectrum.



The gas can be identified by the number of lines produced by the spectra

Level 3 responses must include a discussion of the production of emission spectra and how the spectrum of a gas can be used to determine the elements that the gas contains.

The following example gives a good discussion of the production of light from electrons changing energy levels. Although it does not mention the difference in frequency or wavelength of the light produced in the lines in the emission spectrum there is sufficient in the discussion with respect to colour for the elements in the gas to be identified.

5 Figure 7 shows an example of an emission spectrum of a gas.

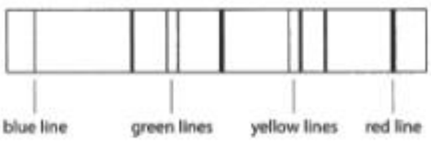


Figure 7

Discuss how emission spectra are produced and can be used to identify the elements in a gas.

(6)

the electrons in the gas will be pushed up a energy level by it being heated up ~~[illegible]~~. And once they go up a energy level it will jump back ~~to~~ down one. because of the attraction ~~of~~ of the nuclear charge. Once the election jumps down to its original energy level. It will release light waves. These light waves will be diffracted by making it the light shine through slits. Once the light wave is diffracted there will be range of different colours shown. This spectrum of ~~diff~~ light colours will then be matched with the emission spectrum of other elements. And if the emission spectrum of the element is similar to the emission spectrum of the gas. That means that gas contains the element you are comparing it to.

In the Level 3 example below the identification of elements in the gas is done first and the difference in frequencies is mentioned. The production of the emission spectrum is then discussed, and the diagram helps by showing the emission of a photon.

5 Figure 7 shows an example of an emission spectrum of a gas.

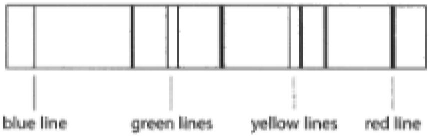


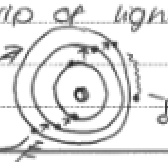
Figure 7

Discuss how emission spectra are produced and can be used to identify the elements in a gas.

(6)

- All different gases (elements in gases) emit light at different frequencies
- this is how they are used to identify each one, because, not one emits light (the exact same amount of light) at the ~~exit~~ specific points of others gases or elements.

- Light is emitted when the electrons in a gas/element becomes excited by specific wavelengths of light they take in energy - this is where the emission spectrum remains/is black) these excited electrons move out to the furthest shell
- when they change back to their resting state they emit one energy and this photon of light, this is where it shows up as a strip of light on the emission spectrum



9

Turn over ▶

Summary

Physics
SECTION C: WAVES IN COMMUNICATION

To improve performance in examinations learners should:

- Learn that antinodes for a stationary wave will be at the points of largest amplitude.
- Practice rearranging equations correctly and always show your working.
- Take care not to repeat the information in the stem of a question.
- Learn to use trigonometrical functions on a calculator.
- Remember that all waves, apart from sound and seismic P waves, are transverse.
- Learn that it is the value of the critical angle in glass that determines if the ray of light in glass will be refracted or totally internally reflected.
- Use the diagrams in questions to give information.
- Draw diagrams to help with explanations.